Final Year Project

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Defense Proposal

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Abstract

Medical Billing in the U.S. healthcare system is slow, complex, and prone to errors. Small clinics and busy practices often rely on manual data entry and coding, which leads to incorrect ICD-10/CPT assignments, frequent claim denials, and delayed payments. This project develops a web-based automated medical billing system that streamlines the entire revenue cycle from patient intake to insurer submission. A key feature is an interactive patient intake module where patients can describe their visit (text or voice); the system uses this input to auto-populate demographic and visit details, reducing initial data entry errors. Doctors then review the record and assign or confirm ICD-10 and CPT codes. The platform applies OCR and NLP to extract clinical information from notes, maps terms to standard codes, and generates CMS-1500/UB-04 claim forms. Before submission, an automated validation engine checks claims against payer rules and an ML model predicts the likelihood of rejection so staff can fix issues proactively. Validated claims are then formatted for electronic submission to insurers. So by combining automation, clinician oversight, and predictive analytics, the system aims to reduce denials, speed reimbursements, and improve billing transparency for providers and patients.

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1. Vision Document

1.1. Introduction

1.1.1. Problem Statement:

Medical billing in the United States is highly complex and mostly manual, leading to frequent coding errors, missing patient details, and high claim rejection rates. These problems cause delays in payments for hospitals, financial losses for clinics, and confusion for patients. There is a need for an automated system that can capture patient information, assist doctors in assigning accurate ICD-10/CPT codes, validate claims before submission, and improve transparency in the billing process.

1.1.2. Objectives

- To develop a web-based system that automates the medical billing process.
- To capture patient details during intake and reduce manual entry errors.
- To assist doctors in assigning accurate ICD-10 and CPT codes.
- To generate and validate claim forms before submission to insurance companies.
- To use machine learning for predicting claim acceptance or rejection.
- To provide dashboards for billing staff and patients for better tracking and transparency.

1.2. Motivation

The complex nature of medical billing system often overwhelms both healthcare staff and patients. The errors in coding and claim submission cause unnecessary delays and financial stress. Our motivation is to apply modern technologies and automation to simplify this process, reduce the burden on clinics, and make billing more transparent for patients.

1.3. Overview

Our proposed system is a web-based medical billing platform which is designed to handle the complete process from the intake of patient details to claim submission. Patients will provide their details during intake, which are recorded in the system. Doctors then confirm and assign ICD-10 and CPT codes, while the system automatically generates claim forms and validates them against insurance rules. A machine learning module predicts the chances of rejection, allowing errors to be corrected before submission. The platform also offers dashboards for staff and patients to track claims and view billing details in a clear and structured way.

1.4. Functionality

The proposed system provides end-to-end automation of the medical billing process, covering patient intake, coding, claim generation, validation, and submission. Each module is designed to reduce manual effort, improve accuracy, and ensure transparency for both healthcare providers and patients.

The main functionalities of the system are described below:

1.4.1. Patient Intake Module

Patients will provide their personal and medical details through an easy-to-use interface. This reduces manual errors and ensures that all necessary data is captured at the start of the billing process.

1.4.2. OCR and NLP Processing

The system will process uploaded medical notes or reports using OCR and NLP to extract key medical information. This feature helps in reducing manual effort and improves coding accuracy.

1.4.3. Coder's Coding Support

Coders can directly assign or validate ICD-10 and CPT codes for each patient record. This ensures that coding remains accurate and aligned with medical standards.

1.4.4. Code Recommendation System

Based on the details entered by the patient, the system will recommend suitable ICD-10 and CPT codes. The doctor will then review and confirm or modify these recommendations before final submission.

1.4.5. Claim Form Generator

The system will automatically generate claim forms such as CMS-1500 and UB-04 with all the required patient, provider, and coding information. This ensures standardization and saves time for billing staff.

1.4.6. Validation Engine and Claim Prediction

Before submission, the system will validate claims against insurance rules and guidelines. Any missing or incorrect information will be flagged for correction. Then we will use a machine learning model that will analyze claims and predict the likelihood of acceptance or rejection. This allows staff to resolve potential issues before claims are submitted.

1.4.7. Dashboard

The dashboard will provide billing staff with tools to track submitted, pending, and rejected claims. Patients will also have access to their bills in a clear and transparent format.

1.4.8. Notifications

The system will send alerts and updates to both staff and patients about claim status changes, such as submission, approval, or rejection.

2. Tools and Technologies

2.1. Backend (Python with Flask)

Python will be used as the core programming language due to its simplicity and rich ecosystem. Flask will serve as the backend framework for building REST APIs and managing business logic.

2.2. Frontend(React.js)

The user interface will be built using React.js to ensure a responsive and user-friendly experience. This will allow patients, doctors, and billing staff to interact with the system through a web browser.

2.3. Database (PostgreSQL)

A relational database such as PostgreSQL will store patient records, billing data, and claim history. PostgreSQL is preferred for its support of structured and semi-structured data formats.

2.4. Optical Character Recognition (Tesseract / Google Vision API)

Tesseract OCR or Google Vision API will be used to extract text

from scanned medical notes and reports. This feature will help in automating the intake of handwritten or printed documents.

2.5. Natural Language Processing (Spacy/NLTK with Medical Libraries)

NLP tools such as SpaCy or NLTK, along with medical term dictionaries, will be used to process text and identify relevant diagnoses and procedures. This will improve accuracy in mapping medical terms to ICD-10 and CPT codes.

2.6. Machine Learning (Scikit-learn/PyTorch/Tensor)

ML frameworks will be used to build and train models for claim rejection prediction. These models will analyze historical claim data to identify risk factors and provide real-time feedback.

2.7. Visual Studio Code

Visual Studio Code is used in our project for writing and managing backend (Python/Django/Flask) and frontend (React.js) code. With built-in debugging, Git integration, and extensions, VS Code streamlines the development process and improves productivity.

2.8. Deployment (AWS)

The system will be deployed on a cloud platform such as AWS to ensure scalability, availability, and security. Docker and Kubernetes may also be used for containerization and efficient resource management.

3. Required hardware and Software resources

3.1. Hardware

Following development PC is required to develop this project

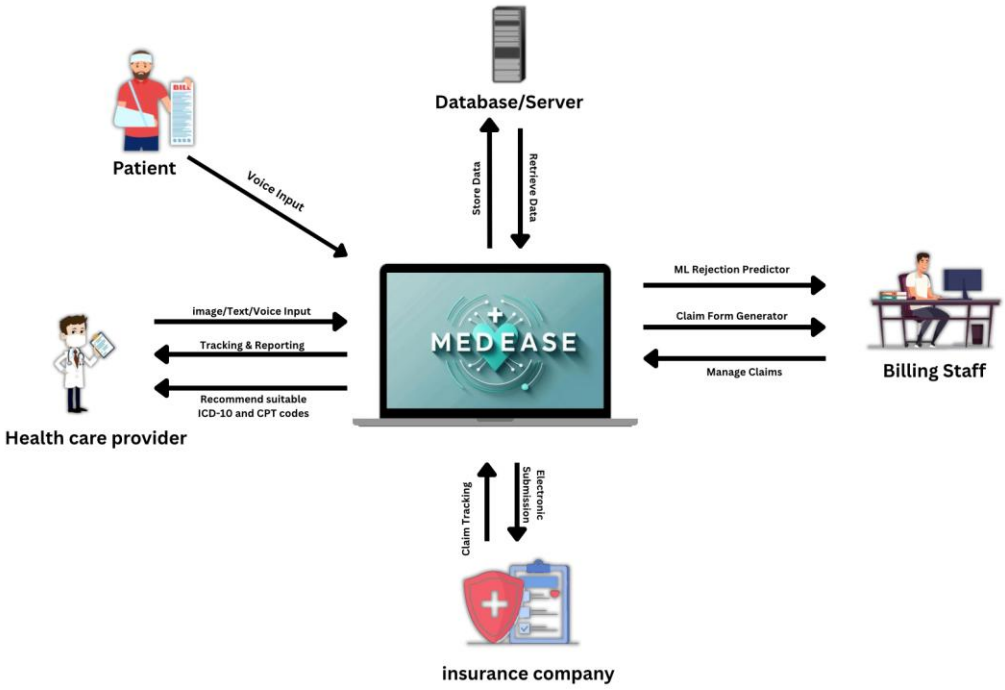
Hardware	Specifications
Development PC	Processor: Intel Core i7 / Xeon (multi-core) RAM: 16 GB minimum (32 GB recommended for ML models) Storage: 500 GB SSD (expandable for patient data storage) 1280 x 800 minimum screen resolution.

3.2. Software

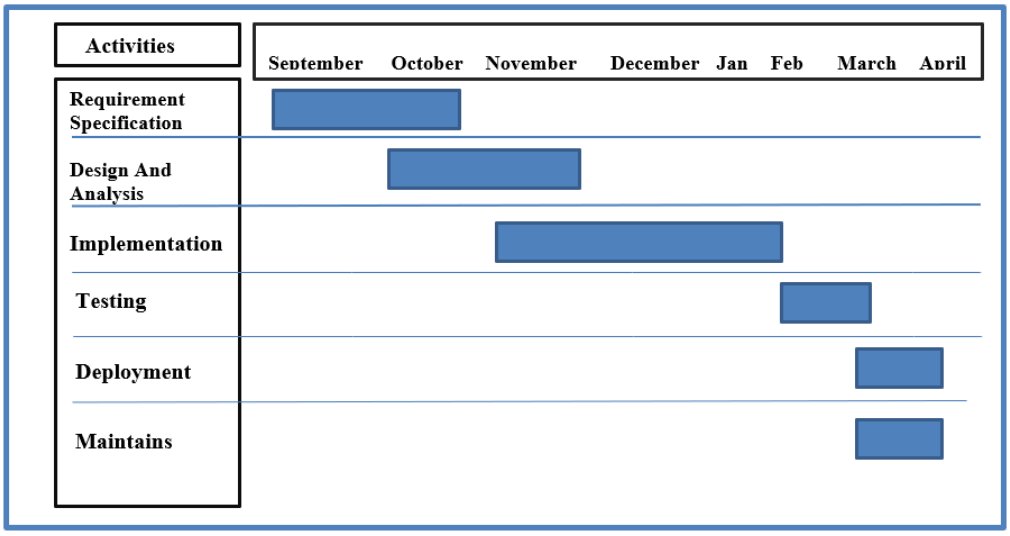
Following softwares are required for the development of this project

Software	Purpose
VS Code	Python Django/Flask,React.js
Postman	API Testing
Microsoft Visio	UML Designing
Python-Tesseract	OCR Tesseract
Jupyter Notebook	Spacy, NLTK ,NLP processing,Scikit-learn ,PyTorch
Docker	AWS Deployment
pgAdmin	PostgreSQL

4. High Level Diagram



5. Timeline



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